

JKU Visual Data Science Lab

AI and Visualization Lab WS26

VIS4AI Phase I Practical Examples

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23.10.2026

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VIS4AI

AI4VIS

VIS4AI

VIS4AI

Assignment

Hands-on experience using visualization to explain AI models and diagnose their failure modes.

We are going to answer two questions:

Phase I How does the model behave?

Phase II What did the model learn?

Assignment > Phase I How does the model behave?

From last session

Design a **visualization** that **helps you observe patterns** in what goes in and what comes out.

Use visualization as an investigative tool to explore patterns in model behavior

What patterns should I look for?

- Errors/confidence change
- Biases
- Hallucination
- Safety Violation
- Shift in output characteristics
- …

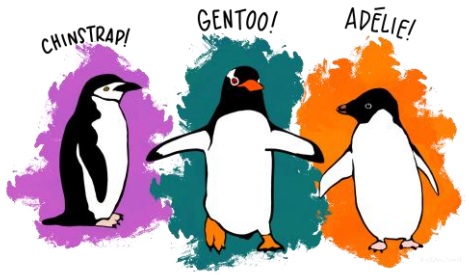
What input data should I use?

- An official test set
- An established benchmark for the task
- A synthetic dataset
- The training dataset

Scatter Plots as Investigative Tool

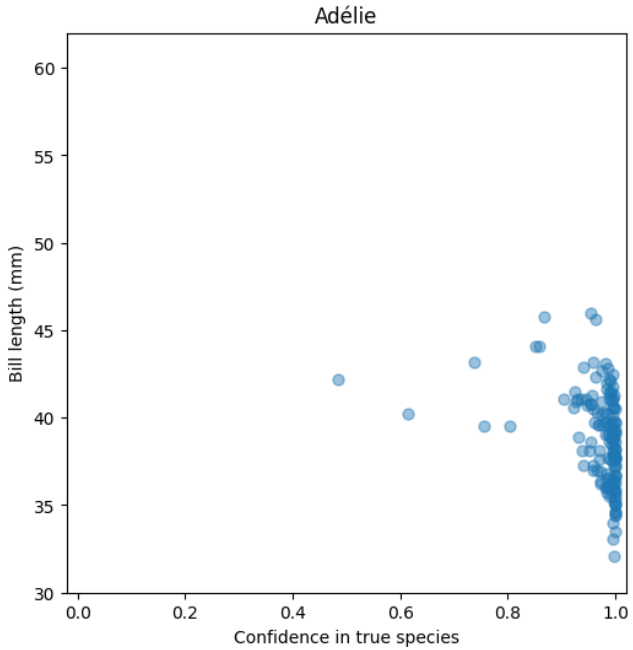
Data item:

	Target			
Bill length	Species	Sex	Island	Model Confidence
Y-Position				X-position



How to include more variables?

Each variable occupies a dedicated visual channel

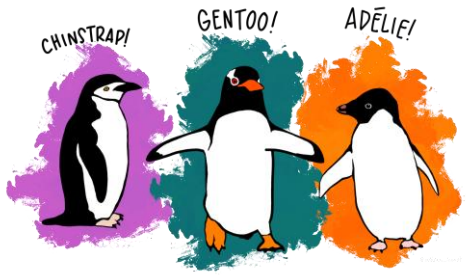


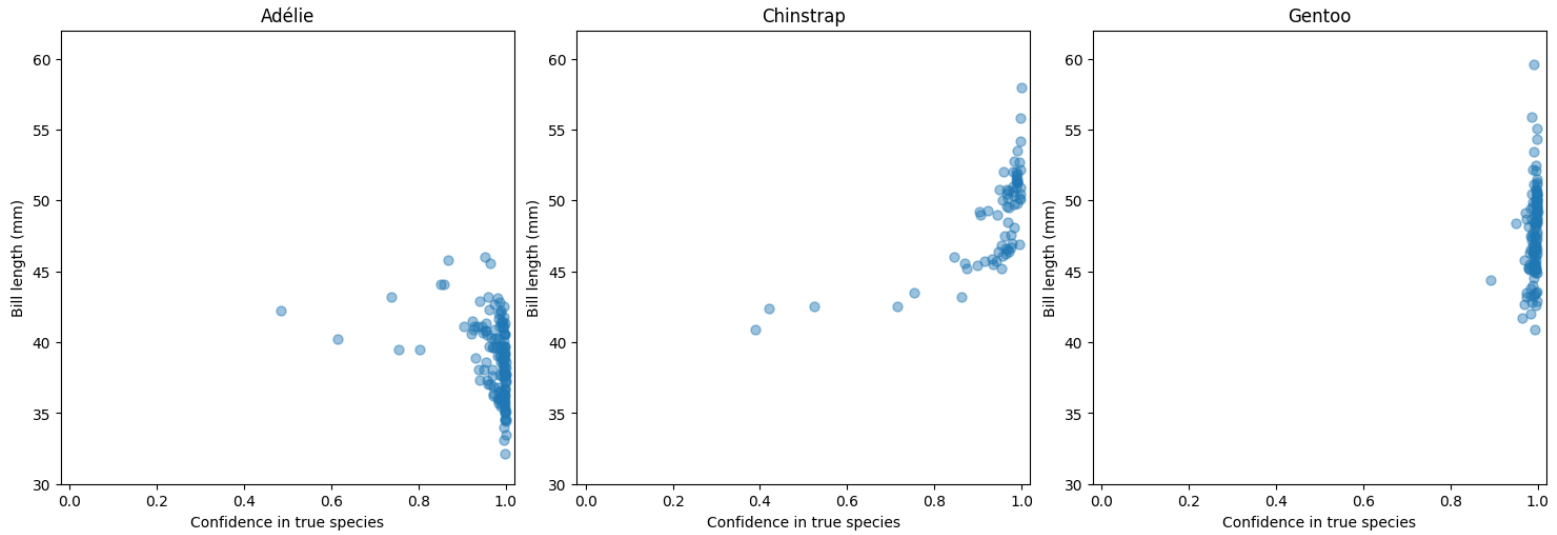
Scatter Plots as Investigative Tool

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Data item:

	Target			
Bill length	Species	Sex	Island	Model Confidence
Y-Position	Small Multiples			X-position





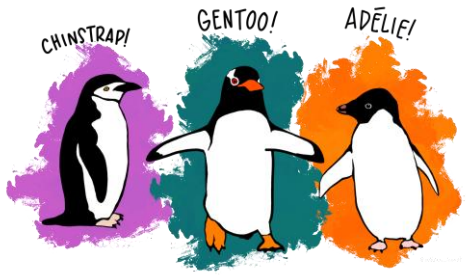
Scatter Plots as Investigative Tool

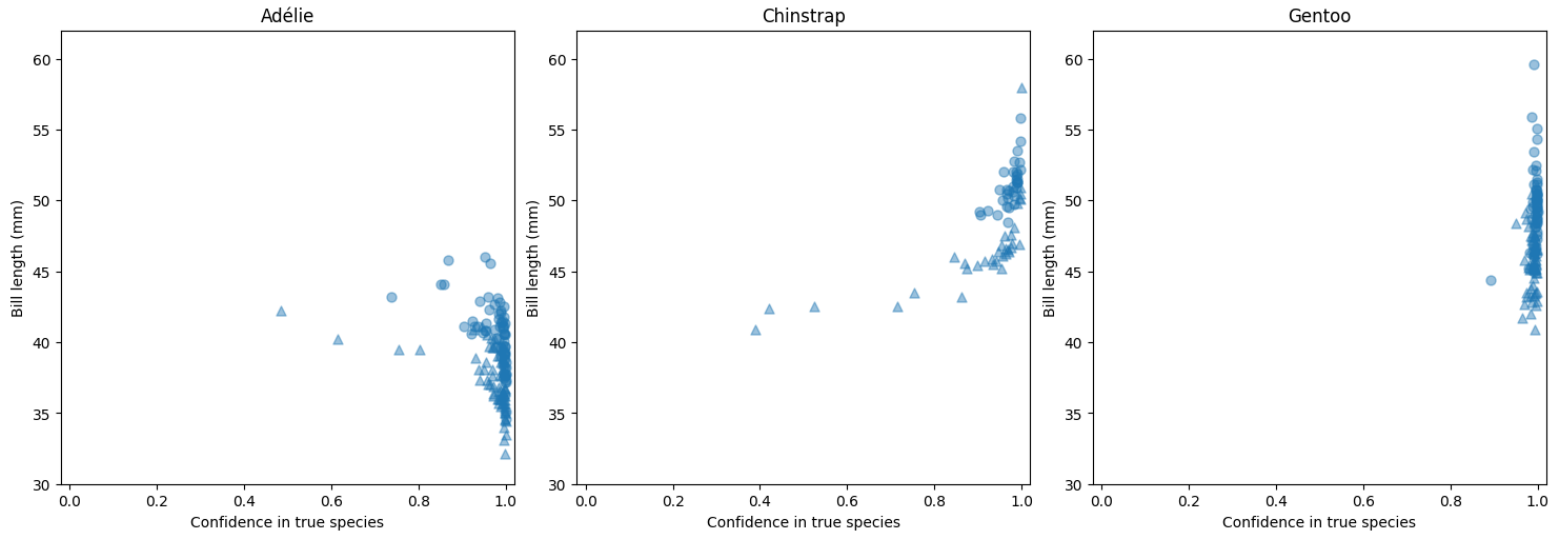
VIS4AI

Target

Data item:

Bill length	Species	Sex	Island	Model Confidence
Y-Position	Small Multiples	Shape ● Male ▲ Female		X-position



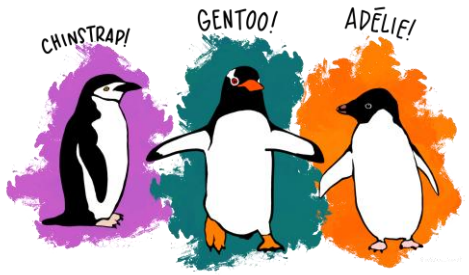


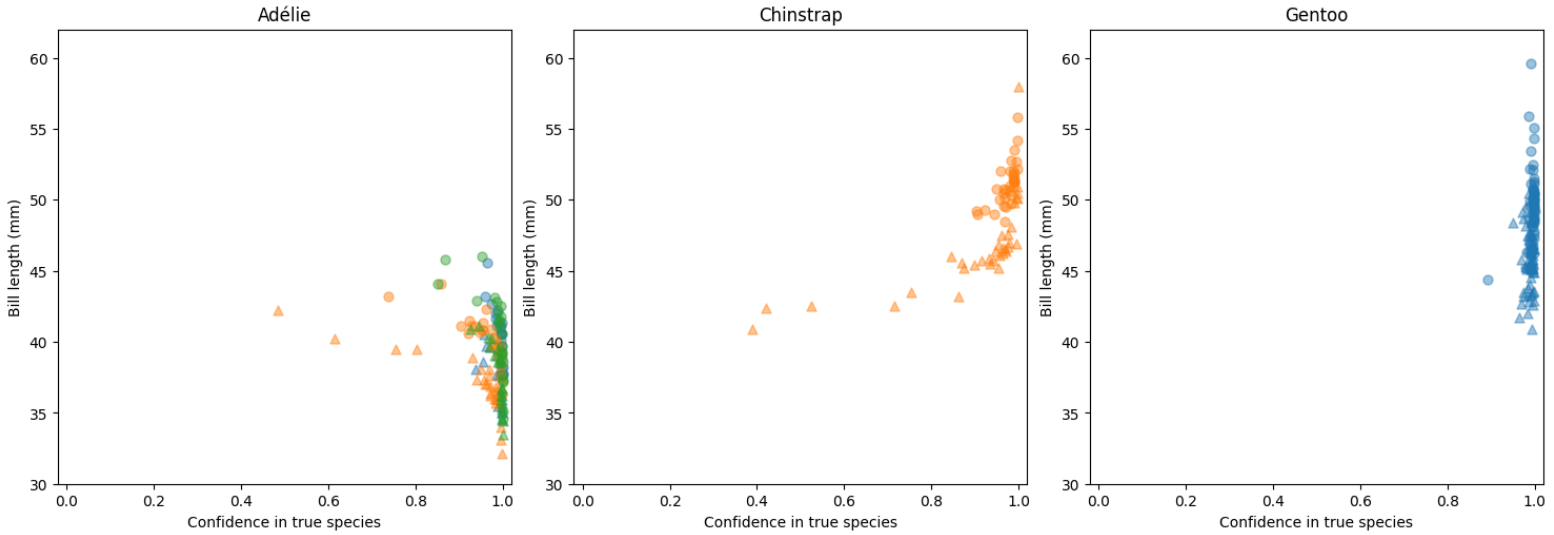
Scatter Plots as Investigative Tool

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Data item:

	Target			
Bill length	Species	Sex	Island	Model Confidence
Y-Position	Small Multiples	Shape Male Female	Color Biscoe Dream Torgersen	X-position





Visual channels are limited

What if we have high-dimensional data?

What if we want to see all in one?

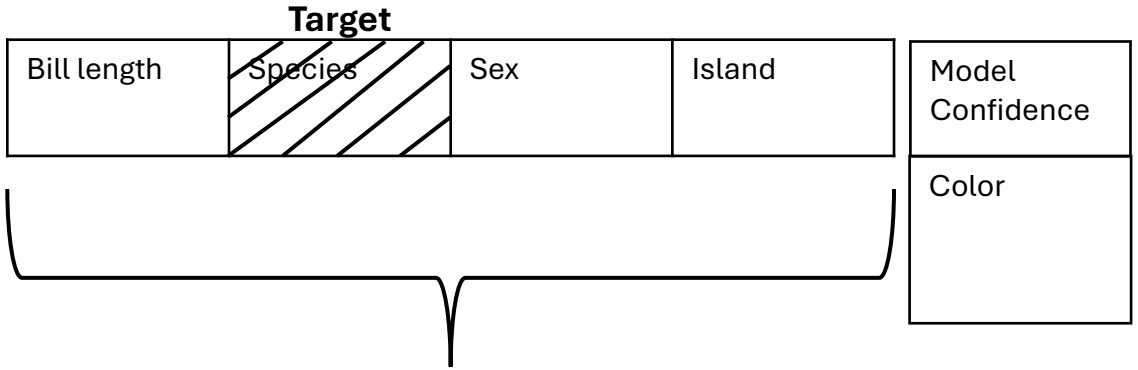
Interpretation

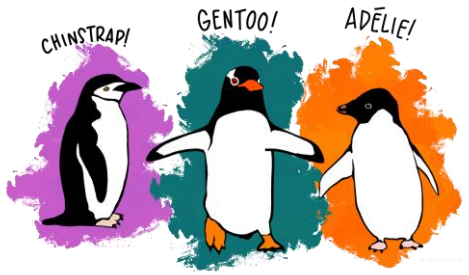
- Longer bills are associated with lower confidence
- In contrast, Longer bills are associated with higher confidence
- Too confident on this species
- Penguins with the lowest confidence are females

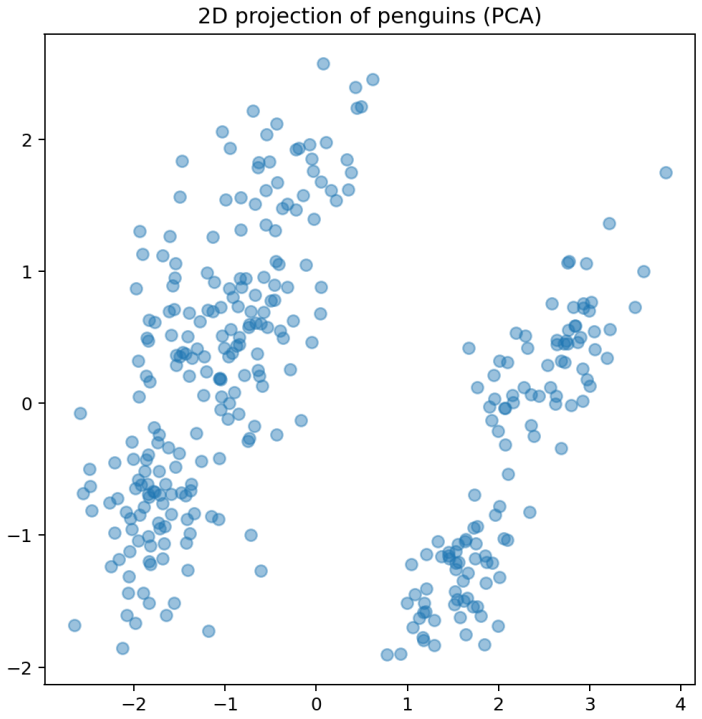
Scatter Plots as Investigative Tool: Projection

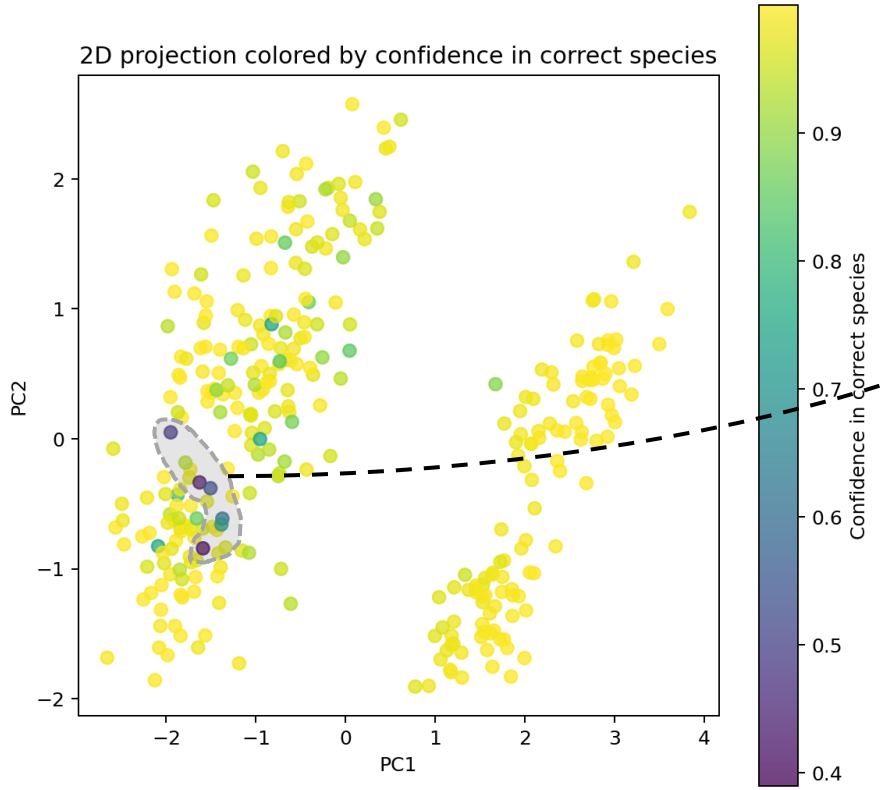
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Data item:







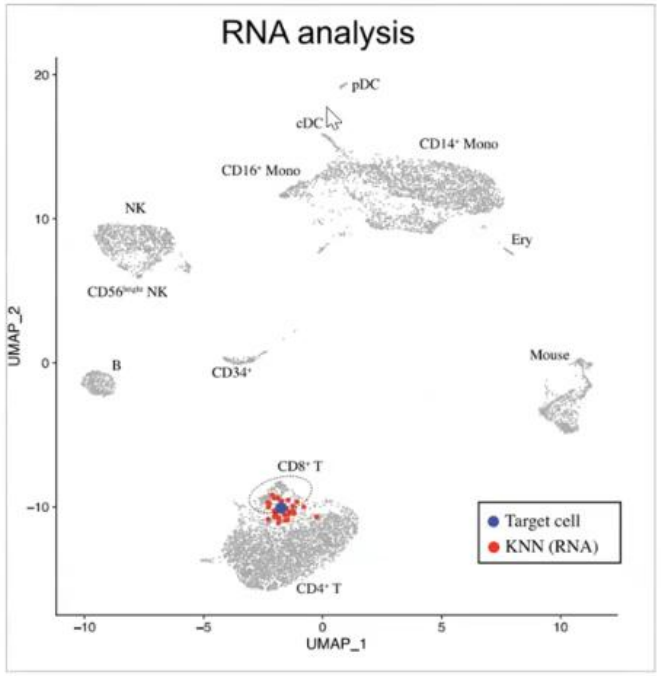


How to interpret it?

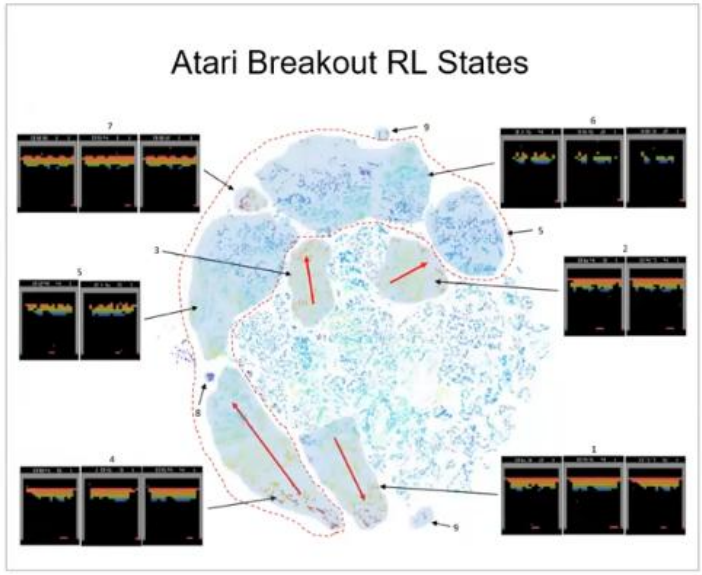
Scatter Plots as Investigative Tool: Projection

VIS4AI

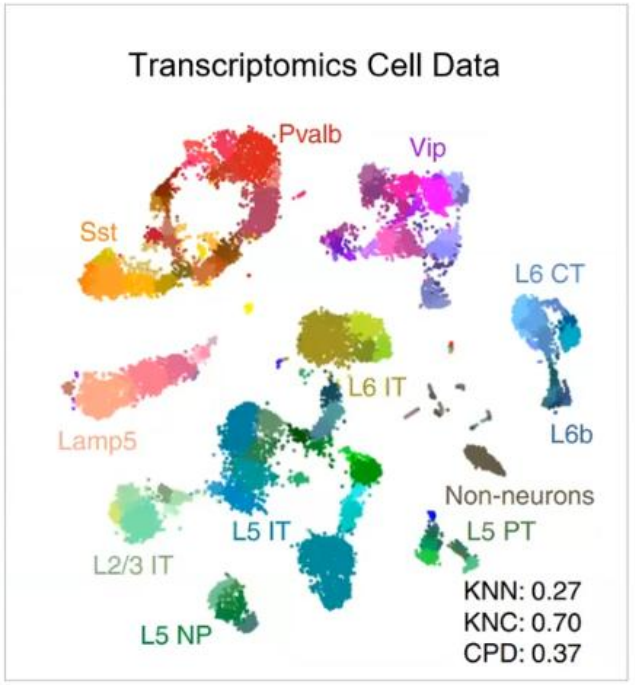
Drill down to the items → Understand → Summarize → Annotate



Hao, Y., Hao, S., Andersen-Nissen, E., Mauck, W. M., Zheng, S., Butler, A., Lee, M. J., Wilk, A. J., Darby, C., Zager, M., et al. (2021).



Zahavy, T., Ben-Zrihem, N., & Mannor, S. (2016).



Kobak, D., & Berens, P. (2019).

Projection Spaces Explorer might help you make sense of your projections 😉

Hinterreiter 2021, Eckelt et al. 2023, Steinparz 2026

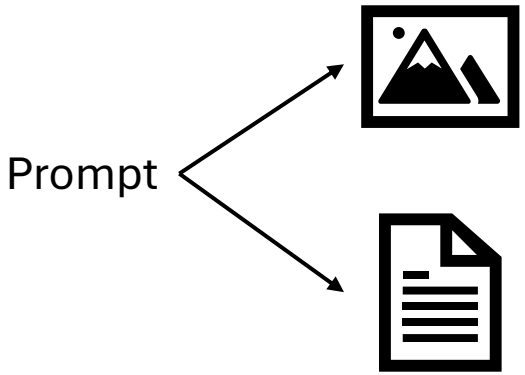
Examples by Christian Steinparz

Scatter Plots as Investigative Tool: Projection

VIS4AI

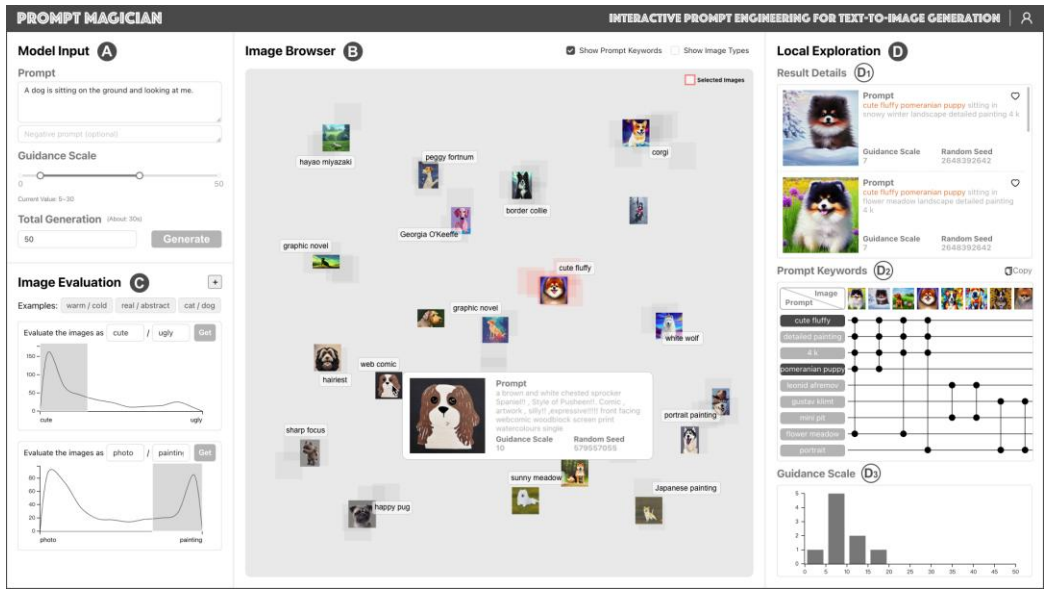
What if my output is richer than my input?

Example



You can still project the input if this serves your analysis.

But you can also project the output! And maybe map the input, e.g., prompt variation.



[PromptMagician]

Yingchaojie et al. 2024

Scatter Plots as Investigative Tool: Projection

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My data items have no attributes like the penguins 😳

You can use a separate encoder (e.g., https://www.sbert.net/)

Pinguins have interpretable attributes, my high-dimensional embeddings won't 😟

If you are interested in a specific behavior, you may be able to derive interpretable attributes related to it.

Example methods to derive interpretable attributes:

LLMs or Traditional NLP to extract linguistic features (e.g., sentiment, negation) check out:

Visual Fingerprints for LLM Generation Comparison, Alnouri et al. 2026

Captioning models; object detectors

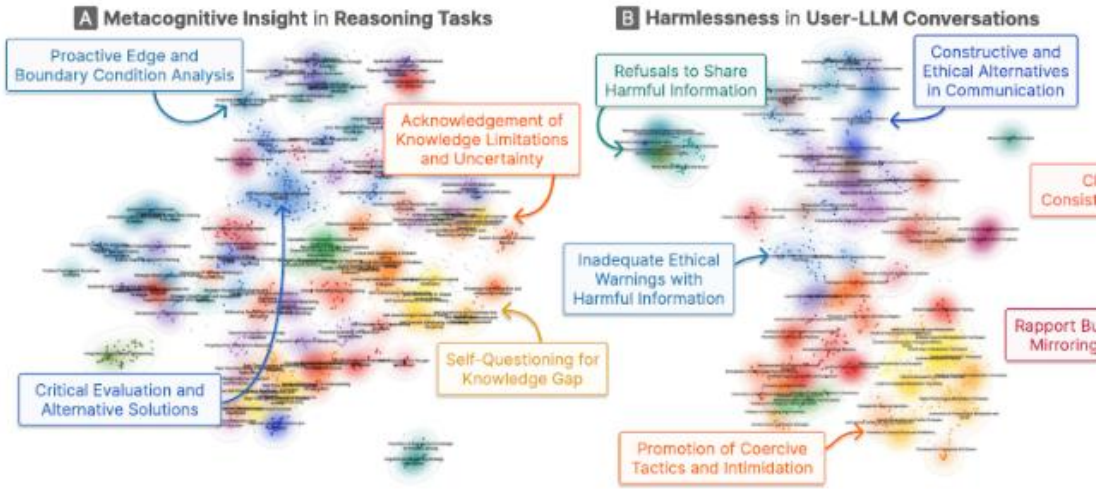
Criteria using LLM-as-a-judge

Diversity metrics

Prompt alignment metrics

...

You can either map these attributes to your high-dimensional embeddings projection or even project your data items using them!



Evalet

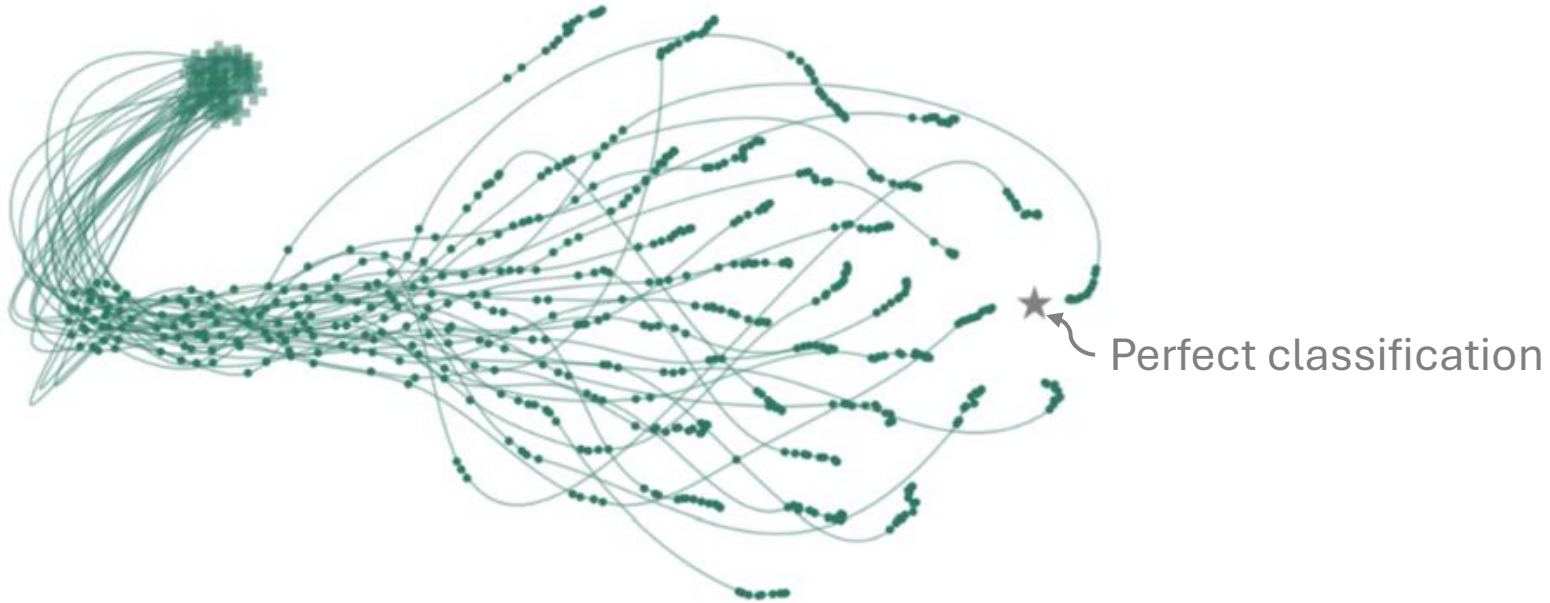
Kim et al. 2026

Scatter Plots as Investigative Tool: Projection

VIS4AI

When your data items form sequences (e.g., checkpoints, prompt edits, conversation turns), you can connect consecutive states into a path in the projection; the path shape shows how behavior evolves.

Time curves: Folding time to visualize patterns of temporal evolution in data
Bach et al. 2015



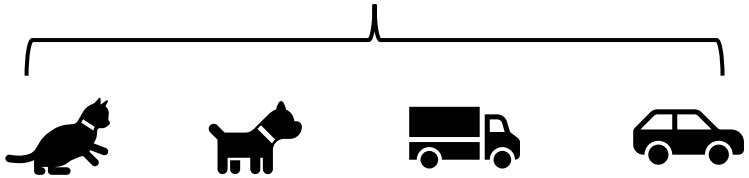
Confusion matrices for the test dataset for a neural network trained on MNIST. Thirty paths are plotted, each path illustrates a different random initialization. Each point illustrates the confusion matrix at a specific training epoch.

Projection Path Explorer: Exploring Visual Patterns in Projected Decision-Making Paths
Hinterreiter et al. 2021

Confusion Matrix as Investigative Tool

VIS4AI

Two classifiers trained on recognizing 4 different objects



Same accuracy. But is one of them better than the other? 🤔

Weaker understanding of the world

	Cat	Dog	Car	Truck
Cat	80	0	0	20
Dog	0	80	20	0
Car	0	20	80	0
Truck	20	0	0	80

Classifier #1

	Cat	Dog	Car	Truck
Cat	80	20	0	0
Dog	20	80	0	0
Car	0	0	80	20
Truck	0	0	20	80

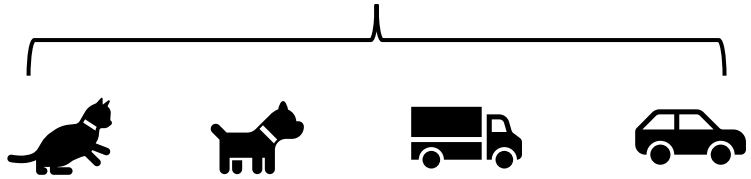
Classifier #2

Learnt something correct about the world!

Confusion Matrix as Investigative Tool

VIS4AI

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	Cat	Dog	Car	Truck
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Dog	0	80	20	0
Car	0	20	80	0
Truck	20	0	0	80

Classifier #1

		Cat	Dog	Car	Truck
Object	Animal	**Cat**	80	20	0	0
		Dog	20	80	0	0
	Vehicle	**Car**	0	0	80	20
		Truck	0	0	20	80

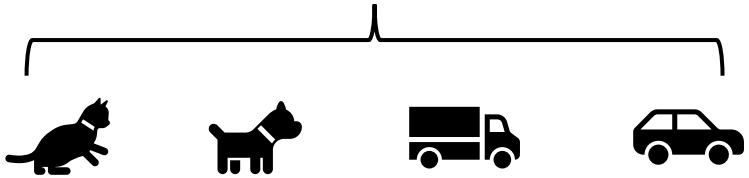
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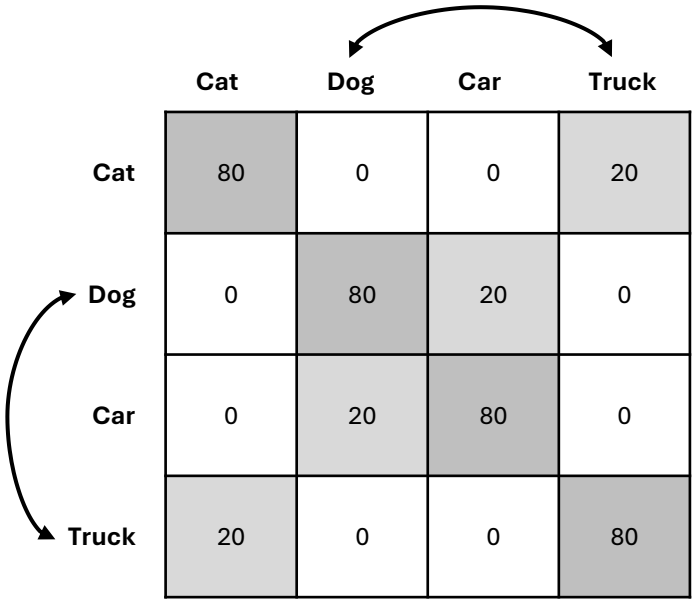
Confusion Matrix as Investigative Tool

VIS4AI

Two classifiers trained on recognizing 4 different objects



To reveal blocks, reorder rows and columns based on error similarities

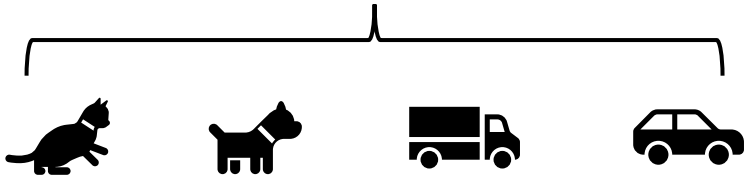


	Cat	Dog	Car	Truck
Cat	80	0	0	20
Dog	0	80	20	0
Car	0	20	80	0
Truck	20	0	0	80

Confusion Matrix as Investigative Tool

VIS4AI

Two classifiers trained on recognizing 4 different objects



To reveal blocks, reorder rows and columns based on error similarities

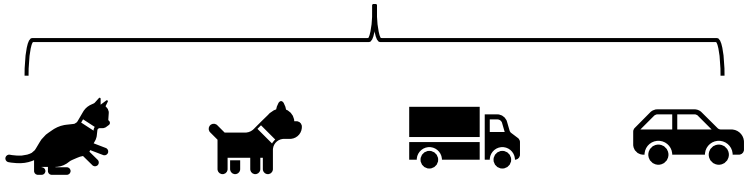
	Cat	Truck	Car	Dog
Cat	80	20	0	0
Truck	20	80	0	0
Car	0	0	80	20
Dog	0	0	20	80

No meaningful hierarchy

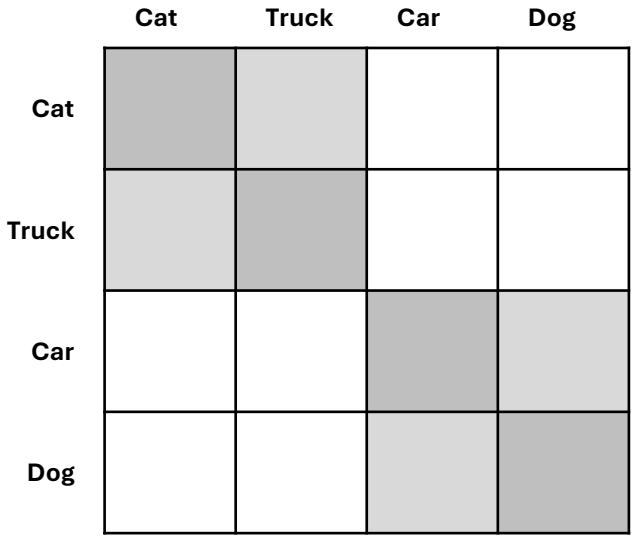
Confusion Matrix as Investigative Tool

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Two classifiers trained on recognizing 4 different objects



To reveal blocks, reorder rows and columns based on error similarities

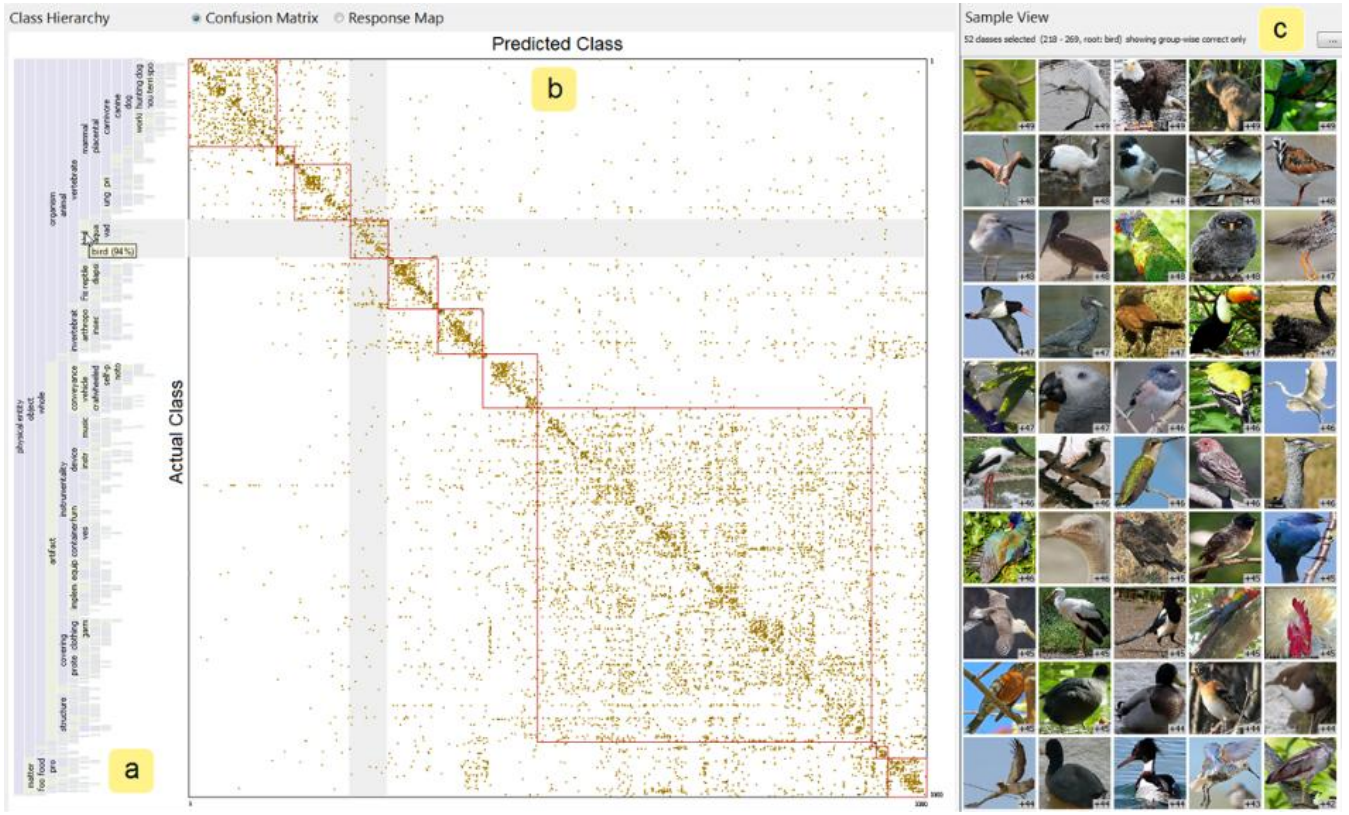


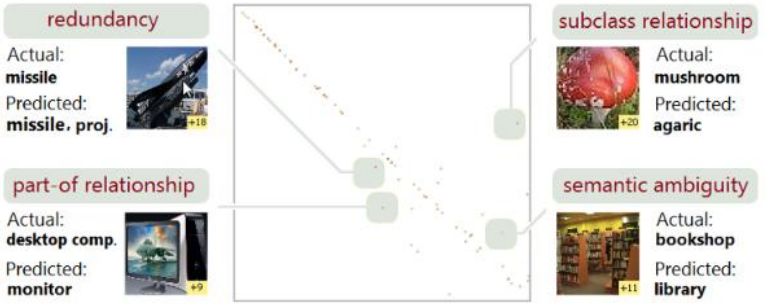
With four classes you can read the numbers. With a thousand classes you can only see patterns.

Confusion Matrix as Investigative Tool

VIS4AI

Drill down to the items → Understand → Summarize → Annotate





Do Convolutional Neural Networks Learn Class Hierarchy?
Alsallakh et al. 2017

Confusion Matrix as Investigative Tool

VIS4AI

If your model is very accurate, your confusion matrix will be sparse. You can still discover interesting structures from your classifier probabilities. Check out:

Visualizing Classification Structure of Large-Scale Classifiers
Alsallakh et al. 2020

If you're interested in exploring how well your model's behavior aligns with real-world knowledge, beyond confusion matrices and classification tasks, check out:

Abstraction Alignment: Comparing Model-Learned and Human-Encoded Conceptual Relationships
Boggust et al. 2025

If you're interested in more exciting confusion metrices, check out:

ConfusionFlow: A Model-Agnostic Visualization for Temporal Analysis of Classifier Confusion
Hinterreiter et al. 2022

InstanceFlow: Visualizing the Evolution of Classifier Confusion at the Instance Level
Pühringer et al. 2020

A Broad Design Space

VIS4AI

These are only examples. The design space is too broad!

Workshop on Visualization for AI Explainability (VISxAI)

IEEE VIS https://visxai.io/

A fresh example on exploring a specific behavior in T2I models

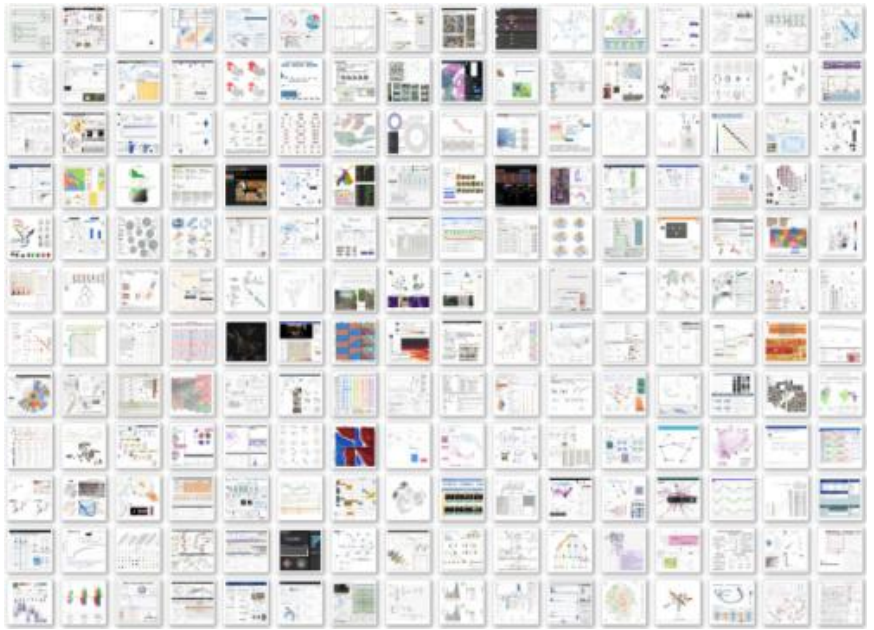
Exploring The Geographic Assumptions in Text-to-Image Models with Underspecified Prompts

Unlu et al. 2026, VISxAI Workshop IEEE VIS

https://t2i-geo-assumptions.jku-vds-lab.at/

You can invent yours too!

Most of the examples we mentioned and more



https://trustmlvis.lnu.se

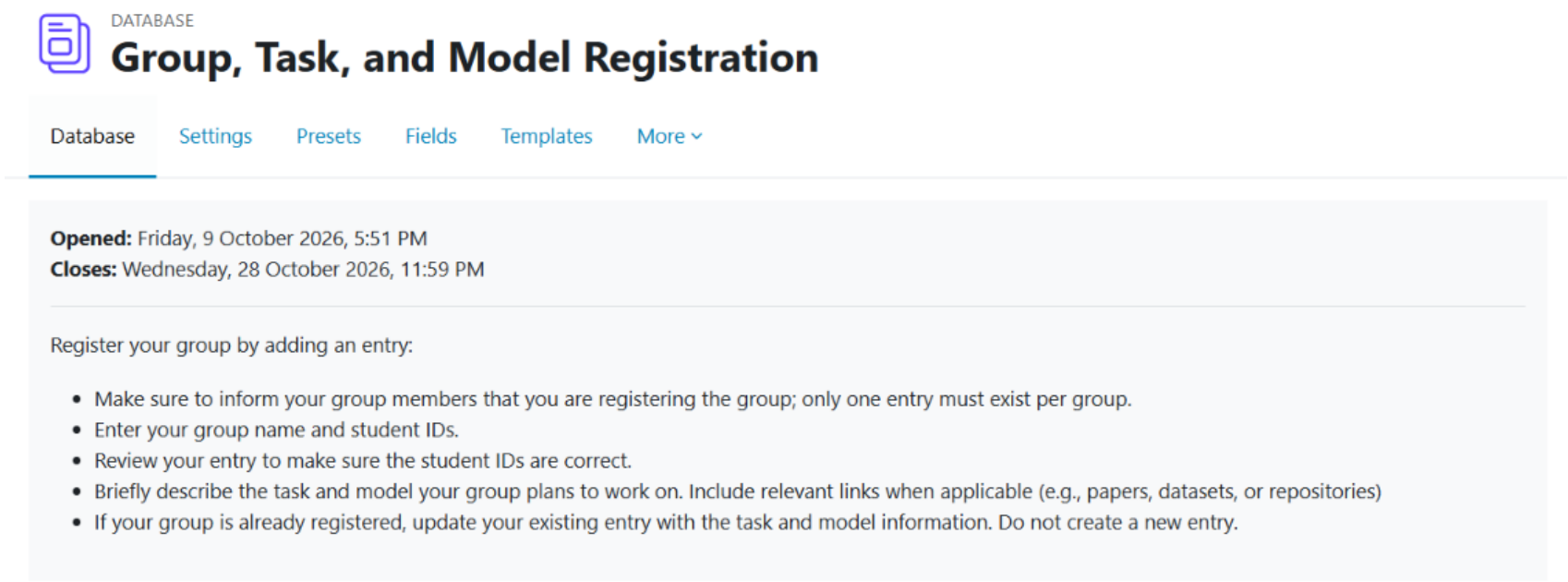
Visualization for Trust in Machine Learning Revisited: The State of the Field in 2023

Chatzimparmpas et al. 2020

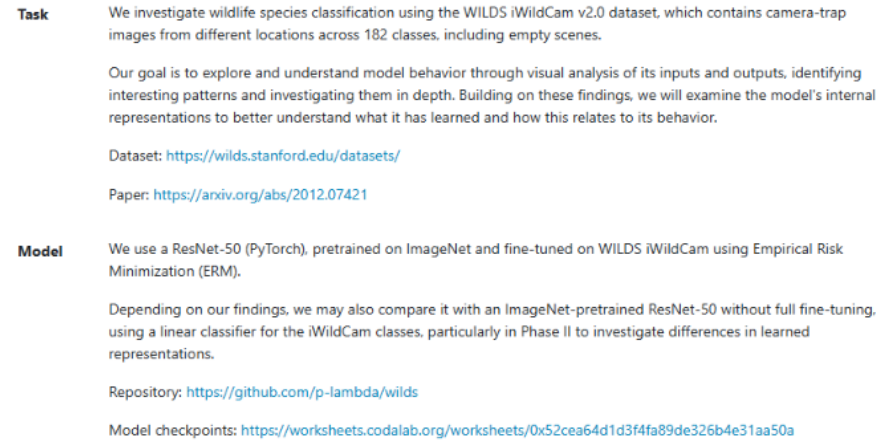
VIS4AI

AI4VIS

Register Your Task and Model on Moodle



Example of an ideal entry by “Lost in Latent Space”



Registration Link

Create Your GitHub Repository

https://github.com



GitHub Repository

Create a **public** repository on GitHub for your code. You will be asked to submit the repository URL.

Include the following files in the root directory of your public GitHub repository, and make sure to fill them out before submitting your project:

- team-info.json
- workload.json

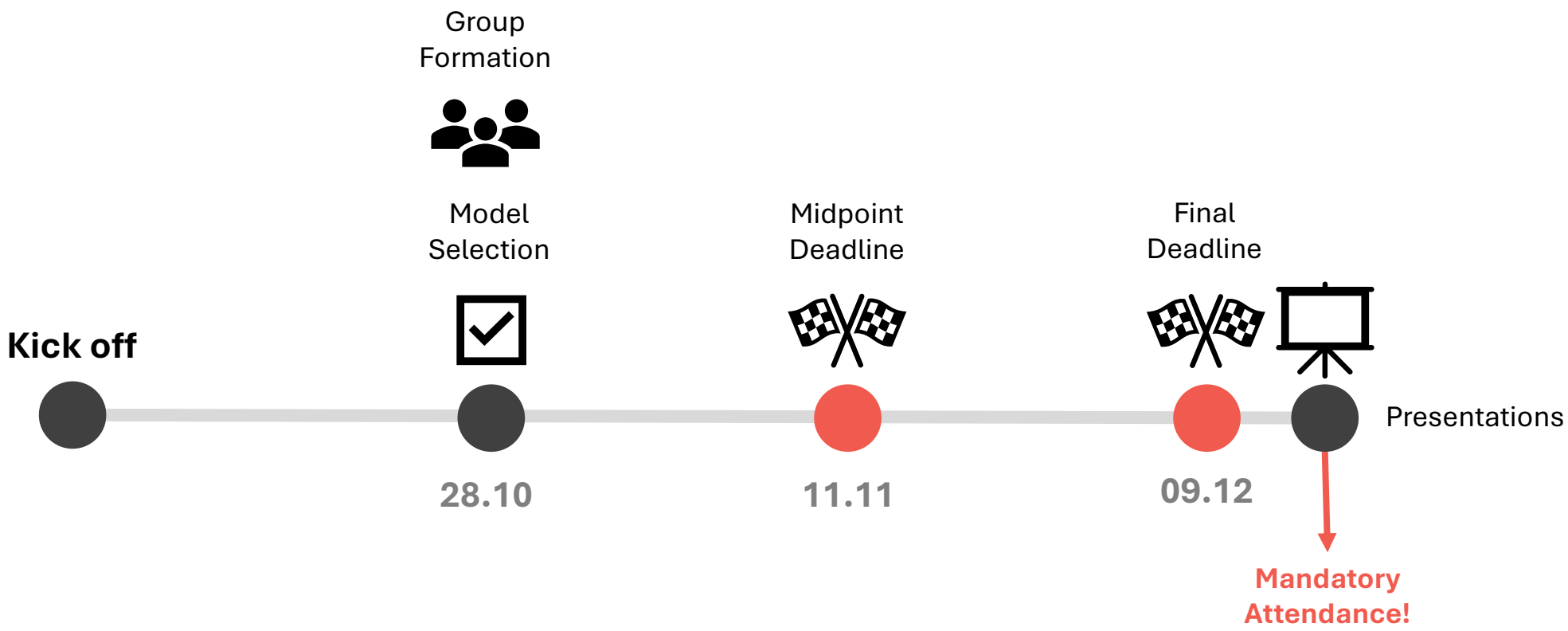
team-info.json

```
{
    "team_name": "Team Name",
    "students": [
        {
            "student_id": "k00000000",
            "first_name": "First",
            "last_name": "Last",
            "email": "first.last@mail.com",
            "github_user": "mygitusername"
        },
        // ...add a record for each student
    ]
}
```

workload.json

```
{
    "team_name": "Team Name",
    "students": [
        {
            "first_name": "First",
            "last_name": "Last",
            "workload": 0.25
        },
        // ...add a record for each student
    ]
}
```

Timeline


Group Formation
Model Selection
Midpoint Deadline
Final Deadline
Kick off
Presentations
28.10
11.11
09.12
Mandatory Attendance!

Next Session

Thu 29.10.26 13:45 - 15:15 S3 055 + online

- Practical examples on AI4VIS

- Organizational details on:
  - Feedback Sessions
  - Midpoint Submission

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Questions